Adarsh Pillai

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Profile

Cybersecurity student focused on **reverse engineering**, with a background in offensive security. Co-founder of the **404squad** CTF team; I organize our near-weekly competitions and mainly solve misc and rev challenges.

I publish detailed CTF writeups and malware analysis, and build and operate **Threat Harbour**, a live SSH honeypot that turns real attacker traffic into public threat intelligence every day.

Education

Placid Vidya Vihar Senior Secondary School — Changanassery, Kerala
Class XII, CBSE (Classes XI–XII) – expected completion Mar. 2027 — *2025 – 2027*

Amrita Vidyalayam — Vazhappally West, Kottayam, Kerala
Class X, CBSE (Classes IX–X) — *2023 – 2025*

Earlier schooling: *Bharatiya Vidya Bhavan Senior Secondary School*, Thirunavaya, Malappuram (Class VIII, 2022–23); *Raja Narayanlal Lahoti English School (RNLES)*, Latur, Maharashtra (Classes I–VII, to 2022)

Experience

Co-Founder & Core Member, 404squad (CTF Team) — 2026 – Present
Competitive Capture The Flag — *Remote, India*

- One of three co-founders of 404squad; run the team's near-weekly competition cycle: scheduling, challenge triage across web/pwn/rev/crypto/forensics, and post-event writeup reviews.
- Primary solver on misc and reverse engineering; cross-train teammates and keep a shared methodology so results repeat.
- Write up every competition publicly so teammates and readers can reproduce each solve step by step.

Independent Security Researcher & Technical Writer — 2021 – Present
Self-directed research, CTFs and lab work — *Remote*

- Publish **15 detailed CTF writeups** (ReRange, pwnable.kr, TryHackMe, picoCTF) covering reverse engineering, binary exploitation and crypto (including custom-VM reversing and ReRange's only Hard challenge): every step, command and its output, with concept explanations.
- Wrote a **static malware analysis of bendi.py**, a plaintext Python SSH worm pulled from Threat Harbour that recruits VPS hosts into a Komari-based botnet: provenance and hashes, C2 infrastructure, persistence and detection ideas.
- Maintain public lab-note repositories for TryHackMe rooms, picoCTF and TCM's Practical Ethical Hacking: enumeration, exploitation path, and what would have stopped it.

Projects

Threat Harbour | *Cowrie, Python, Oracle Cloud, Grafana/Loki, GitHub Actions* — Aug. 2026 – Present

- Deployed and hardened a **Cowrie 3.0.13 SSH honeypot** on an OCI Free Tier VM (ap-hyderabad-1): single exposed port, NSG rules, restricted read-only SSH operator account, localhost-bound monitoring never exposed publicly.
- Built a **daily 00:00 UTC pipeline** that SSHes to the sensor, runs a stdlib-only parser on-box and commits only aggregates: raw logs, source IPs, payloads and keys never leave the machine; out-of-band OCI API reboot handles a wedged sshd.
- Instrumented the sensor with **Grafana + Loki + Promtail** (SSH-tunnel only) and gated every publish on a fail-loud **pytest** suite with GitHub Actions CI.
- **2,925,425 events / 8,198 unique IPs / 386,289 sessions** analyzed from Aug. 27 to Nov. 4, 2026. Median session 2.3s; 74.8% of commands are discovery and fingerprinting, with repeated persistence attempts against authorized_keys. Findings are written up in the research docs.
- Published a public **intel dashboard** rebuilt daily from the same metrics: full, defanged attacker commands, SHA-256 hashes with VirusTotal/MalwareBazaar lookups, and **IoC feeds** (TXT, CSV, JSON) of hashes, file paths, source networks and credentials.

Security Automation Toolkit | *Python, Bash* Jul. 2026 – Present

- Wrote a set of small CLI utilities for repetitive assessment work: port scanner, subdomain hunter, hash cracker, DNS scanner, host-to-IP resolver, IP tracker and WordPress content discovery.
- Shell tooling for **recon and web application enumeration** (service and version sweeps, content brute-forcing), documented with usage examples for CTF and lab use.

Security Portfolio & Publishing Platform | *Astro, TypeScript, CSS, GitHub Actions* Aug. 2026 – Present

- Built a static security portfolio in **Astro** with a keyboard command palette, full-text search (Pagefind), filterable writeup index, live moon-phase widget and strict CSP (hashed inline script only); no backend.
- Wrote the content pipeline in TypeScript: a sanitising Markdown renderer with syntax highlighting, automatic category tagging, source-grounded difficulty badges, per-challenge file listings and generated share cards.
- Made it discoverable by search engines and AI agents: generated sitemap, schema.org structured data, `llms.txt` and IndexNow pings. GitHub Actions deploys daily, on push and on cross-repo dispatch, with zero manual steps.

Competitive Results

Lun4R CTF 2026 *qualifiers*, 404squad: **#2** overall *proof · certificate*

PwnSec CTF 2026, 404squad: **#6** human division, 237 teams *proof · certificate*

Lun4R CTF 2026 *finals*, 404squad: **#9** overall *proof · certificate*

z0d1ak CTF 2026, 404squad: **#13** human, #75 overall *proof · certificate*

TFC CTF 2026, 404squad: **#23** human, #172 overall *proof · certificate*

H7CTF 2026, 404squad: **#58** overall *proof · certificate*

TryHackMe: top 2% globally, 100+ rooms completed *profile*

ReRange: 5 reversing challenges incl. its only Hard; **pwnable.kr**: 5 challenges (fd, collision, bof, passcode, dragon); **picoCTF**: heap and rev

Technical Skills

Security: Reverse Engineering, Binary Exploitation (pwn), Web Exploitation, Network Security, OSINT, Threat Intelligence, Honeypot Operations
Tools: Burp Suite, ffuf, Nuclei, Nikto, sqlmap, Nmap, Wireshark, Ghidra, Binary Ninja, pwntools, pwndbg, Autopsy, Binwalk
Languages: Python, Bash, C, SQL, Go
Platforms: Linux, Docker, Git, GitHub Actions, Oracle Cloud (OCI), Grafana, Loki, Astro
Currently Learning: VM Reversing, Embedded Security, Malware Analysis, Firmware Analysis

Certificates & Training

Hacker Holidays: TryHackMe, 2026 *verify*

Advent of Cyber 3: TryHackMe, 2021 *verify*

Practical Ethical Hacking: TCM Security, 2021 *verify*